

Introduction

Our team hypothesized that students at the University of North Carolina at Chapel Hill (UNC) who are social drinkers will be more likely to participate in online sports gambling compared to those who do not.

This question is of growing importance amid the national rise in sports betting rates among young adults. As of January 2025, 35% of online sports betting accounts are adults between the ages of 18 and 34 (Statista Research Department, 2024). Many avenues have made sports betting more accessible and ingrained in young adults' lives, including the quantity of mobile apps, promotion of these apps at sporting events, and low levels of government oversight. Young men who participate in sports betting often develop antisocial and obsessive habits with sports betting apps that are mislabeled as dedication in the social space. Furthermore, many young adults are currently facing a weakened job market, making an immediate and large payout seem appealing (Rose-Berman, 2025). Finally, mobile sports betting apps are available 24/7 from the comfort of one's own home. This constant accessibility increases the dangers of potential addiction.

Monetary loss from online sports gambling is not the only worry. The anxiety and stress that a sports bet loss can cause can either motivate future risks or cause players to harm themselves (David et al., 2026). Losing a bet can be embarrassing and cause a bettor to become more vulnerable to the manipulative tactics and elements of sports betting. One such example is the parlay, which is where multiple bets can be looped into one. If the parlay is successful, then it yields a more rewarding payout than a regular bet. However, if one element of the parlay fails, then the entire parlay fails, creating a feeling of "almost but not quite" that can hook users.

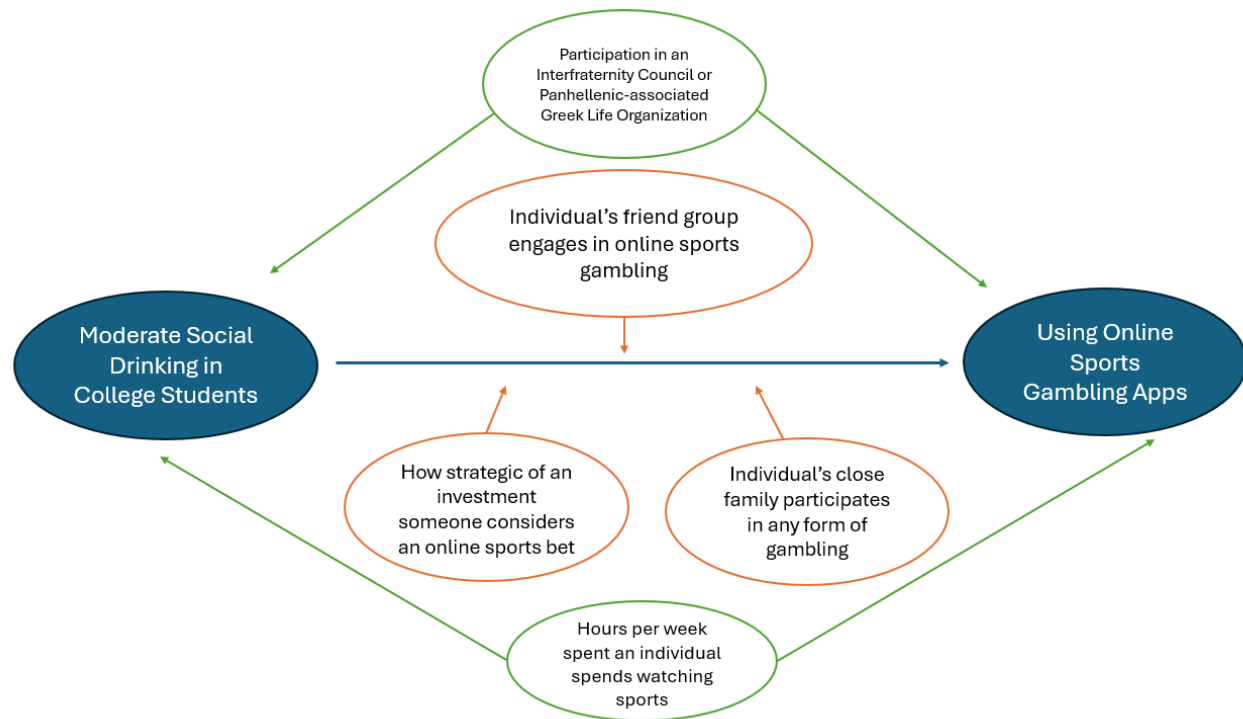
Another example is in-app “bonus bets,” where a large sum of betting money like \$500 is promised to new players who wager a measly \$5 or \$10. Often, this bonus money is eventually lost, but while betting with it, the user becomes desensitized to app usage, resulting in them investing more of their own money. These predatory tactics show that deeper consideration of sports betting and the demographics they target is very important so that proper education and resources can be given to those who most need it (Rose-Berman, 2025).

Methods

We used a cross-sectional survey study design in which participants reported behaviors over the past semester at UNC. Our target population was all UNC undergraduates. We chose to dispense our survey through GroupMe as we felt this would give us a random sample of UNC students. We sent our survey to the following groups: 2025-2026 Trumpet Meme Chat (34 members), 2026 OLs (41 members), UNC BPSS (775 members), BOUNC Cricket GroupMe (164 members), AWM 2025-26 (147 members), ペラペラClub!! (97 members), UNC VTT ‘26 (89 members), and UNC CLUB JUDO (321 members). We received 100 total usable responses before ending our survey, resulting in a response rate of $100/1668 \approx 6\%$. We conducted all statistical analysis, data cleaning, and visualization creation in R version 4.5.3.

In our original survey, to measure our exposure, we asked: “How many alcoholic drinks, on average, did you consume per week this semester? Please note that this survey is anonymous and confidential.” To make this variable binary, we coded answers of 0 drinks per week as 0 and all other answers as 1. In our original survey, to measure our outcome, we asked: “Did you place an online bet on a sporting event from January 1st, 2026, to March 14th, 2026?” This question had the option choices of “Yes” and “No”, which we coded as 1 and 0, respectively.

Figure 1: DAG including Confounders and Effect Measure Modifiers



Our data cleaning was comprised of removing any survey responses that were not completed, removing any of the respondents' personally identifying information, checking data for data entry errors, inconsistencies, or implausible values, editing some survey responses to standardize questions (e.g., "zero" was converted to 0), creating binary variables for exposure, outcome, days of the week they drank, and created a column of sum of days they drank.

Results

Of the 1,688 undergraduates at UNC to whom the survey was sent, there were 100 respondents, resulting in a response rate of approximately 6%. Of these respondents, there was a 71% prevalence of consumption of alcoholic beverages, quantified by the students who drink 1

or more on a weekly basis. There was a 12% prevalence of online sports betting, quantified by the students who had placed at least one online sports bet from January 1 to March 14 of 2026. The results of the survey are detailed in **Table 1**, which is sorted into exposure, outcome, demographics, and other variables. It also splits respondents up by whether or not they met the criteria for the outcome. The prevalence of the exposure aligns with our hypothesis, as a previous study surveying 168,297 students from 78 US campuses recorded a 61% two week period-prevalence of alcohol use (Ranker and Lipson, 2022). The prevalence of the outcome is also similar to previous literature; a study measuring college student sports betting behaviors at the University of Michigan, a U.S. public university similar to UNC Chapel Hill in academic prestige, found that 12.6% of undergraduate respondents placed a sports bet outside of one-off popular events (Mortenson 2026).

Table 1. Descriptive statistics of social drinking, online sports betting, and key potential confounding variables in UNC undergraduate students, Spring 2026 (n = 100)

Sample Characteristics	No Online Sports Betting (N=88)	Online Sports Betting (N=12)	N (%) Mean (SD)
Drinking (exposure)			
Never (0 drinks / week)	28 (32%)	1 (8%)	29 (29%)
Infrequent (1-7 drinks / week)	49 (56%)	8 (67%)	57 (57%)
Frequent (> 7 drinks / week)	11 (12%)	3 (25%)	14 (14%)
Common drinking location			
Your living quarters	13 (15%)	1 (8%)	14 (14%)
A friend's living quarters	23 (26%)	6 (50%)	29 (29%)
A bar	23 (26%)	4 (33%)	27 (27%)
A restaurant	1 (1%)	0	1 (1%)
Does not drink	28 (32%)	1 (8%)	29 (29%)
Common drinking item			
Beer or cider	18 (20%)	6 (50%)	24 (24%)
Wine	12 (14%)	0	12 (12%)
Mixed drinks/cocktails	24 (27%)	2 (17%)	26 (26%)
Liquor (no mixers)	6 (7%)	3 (25%)	9 (9%)
Does not drink	28 (32%)	1 (8%)	29 (29%)
Number of drinking days per week	1.69 (±1.43)	2.83 (±1.70)	1.80 (±1.50)
Number of "black out" side-effects	1.47 (±2.60)	4.18 (±6.59)	1.79 (±3.51)

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Percent of friend group who drinks	70.12 (± 27.77)	80.67 (± 14.41)	71.54 (± 26.77)
Online Sports Betting Occurrences from Jan 1-March 14 (Outcome)			
0 Times	88 (100%)	0	88 (88%)
Infrequently (1-10 Times)	0	8 (67%)	8 (8%)
Frequently (10+ Times)	0	4 (33%)	4 (4%)
Sports Betting Apps on Mobile Device			
0 Apps	83 (94%)	1 (8%)	84 (84%)
1+ Apps	5 (6%)	11 (92%)	16 (16%)
Net Gain/Loss	0	65.45 (± 143.20)	7.27 (± 50.20)
Average Betting Amount	0	1.24 (± 5.59)	11.14 (± 13.61)
Desire to Reduce Online Sports Betting			
Yes	0	1 (8%)	1 (1%)
No	88 (100%)	11 (92%)	99 (99%)
Been Told to Stop Online Sports Betting			
Yes	0	3 (25%)	3 (3%)
No	88 (100%)	9 (75%)	97 (97%)
Demographics			
Household Income			
≤ \$90,000	12 (14%)	1 (8%)	13 (13%)
> \$90,000	72 (82%)	11 (92%)	83 (83%)
Prefer not to say	4 (4%)	0	4 (4%)
Gender			
Female	49 (56%)	1 (8%)	50 (50%)
Male	39 (44%)	11 (92%)	50 (50%)
Greek Life			
Yes	2 (2%)	1 (8%)	3 (3%)
No	86 (98%)	11 (92%)	97 (97%)
Age	20.38 (± 1.20)	20.60 ($\pm .97$)	20.41 (± 1.18)
Other Characteristics			
Gambles			
Yes	9 (10%)	12 (100%)	21 (21%)
No	78 (89%)	0	78 (78%)
Prefer not to say	1 (1%)	0	1 (1%)
Parental Gambling			
Yes	12 (14%)	1 (8%)	13 (13%)
No	76 (86%)	10 (84%)	86 (86%)
Prefer not to say	0	1 (8%)	1 (1%)
Percent of Friends who Sports Bet			
0% - 20%	62 (70%)	4 (33%)	66 (66%)
21-100%	26 (30%)	8 (67%)	34 (34%)
Watches Sports			
Never (0 hrs / week)	36 (41%)	0	36 (36%)
Infrequent (1-5 hrs / week)	43 (49%)	9 (75%)	54 (54%)
Frequent (≥ 6 hrs / week)	9 (10%)	3 (25%)	12 (12%)
Opinion on sports betting (0 = pure chance, 100 = completely strategic)	39.55 (± 21.60)	47.67 (± 28.52)	40.67 (± 22.65)

Opinion on sports betting at UNC			
Disagree	49 (56%)	9 (75%)	58 (58%)
Neither agree nor disagree	20 (23%)	0	20 (20%)
Agree	19 (21%)	3 (25%)	22 (22%)

To determine if cofactors were EMMs or confounders, we calculated stratum-specific odds ratios and then compared them to the crude odds ratio. Precise but inaccurate stratified odds ratios implied Confounder, while imprecise stratified odds ratios on opposite sides of the crude odds ratio implied that the variable was an effect measure modifier. If the stratified odds ratios were imprecise and inaccurate with both odds ratios being less than or both greater than the crude odds, they were classified as possibly being a confounder and effect measure modifier. When a covariate was a confounder and effect measure modifier we treated it like an effect measure modifier in the final analysis.

Table 2: Covariate Classification

Covariates	Crude OR	Stratum 1 Prevalence Odds Ratio and 95%CI	Stratum 2 Prevalence Odds Ratio and 95%CI	Compare to Stratification Table. Which Row does it look like
Survey Respondent Gambles or Not (Yes/No)	5.133	1.125 (0.060 - 21.087)	2.118 (0.092 - 48.628)	Confounder
Gender (Woman/Man)	5.133	0.800 (0.025 - 25.222)	5.600 (0.648 - 48.424)	Effect Measure Modification
Does The Survey Respondent's Parents Gambles? (Yes/No)	5.133	1.000 (0.027 - 36.623)	4.154 (0.498 - 34.670)	Effect Measure Modification/Confounder
Opinion On If People At UNC Sports Bet (Agree/Disagree)	5.133	0.533 (0.038 - 7.487)	11.400 (0.624 - 208.258)	Effect Measure Modification
Hours Per Week Watching Sports? (1-5 Hours/6+ Hours)	5.133	1.818 (0.195 - 16.944)	8.000 (0.329 - 194.422)	Effect Measure Modification
Household Income (\$15,001 - \$90,000/\$90,000+)	5.133	2.000 (0.056 - 71.632)	3.846 (0.462 - 32.021)	Confounder
Percentage of Friends Who Sports Bet (0-20% / 20-100%)	5.133	4.400 (0.222 - 87.115)	2.100 (0.214 - 20.640)	Effect Measure Modification/Confounder
View On Sports Betting Between Pure Chance and Completely Strategic	5.133	2.742 (0.296 - 25.424)	4.154 (0.210 - 82.018)	Effect Measure Modification/Confounder

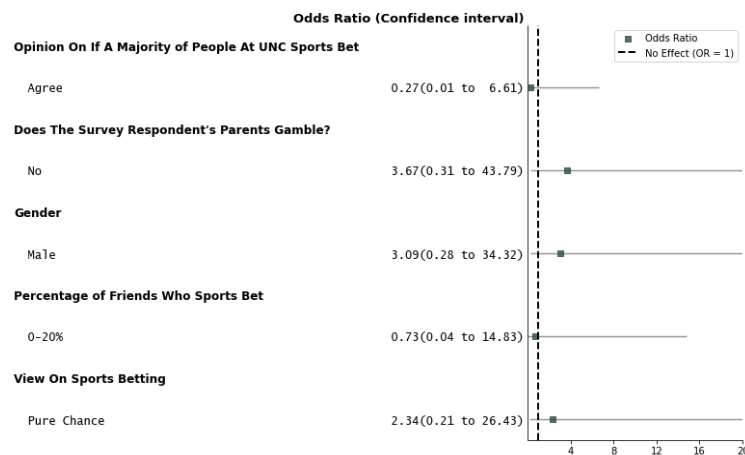
To assess whether our covariates were statistically significant, we compared the crude prevalence odds ratio to the adjusted prevalence odds ratio for each variable, outlined in **Table 3**.

Table 3. Association between drinking and sports betting among UNC undergraduate students using log-odds from logistic model, January - March 2026, n=100

Variable	Response	Crude POR	95% CI	Adjusted POR**	95% CI
Consumption of Alcohol	1+ Drink/Week	5.12	0.63 – 41.56	2.79	0.26 – 30.06
Gender	Woman	100.54	N/A*	0.39	N/A*
	Man	5.60	0.65 – 48.42	3.09	0.28 – 34.32
Parantel Gambling	Yes	0.25	0.03 – 2.24	1.00	N/A*
	No	0.53	0.04 – 7.49	3.67	0.31 – 43.79
Perception of Sports Betting at UNC	Majority sports bet	0.25	0.03 – 2.24	0.27	0.01 – 6.61
	Majority do not sports bet	0.53	0.04 – 7.49	1305.04	N/A*
View on Sports Betting	Strategic Investment	1061.84	N/A*	0.38	N/A*
	Pure Chance	2.74	0.30 – 25.42	2.34	0.21 – 26.43
Percentage of Friends who Sports Bet	0–20%	2.10	0.21 – 25.42	0.73	0.04 – 14.83
	20–100%	844.6	N/A*	895.35	N/A*

*Model broke **Adjusted by Household Income and If the Respondent Gambles

Table 4. Forest Plot Of Adjusted 95% Confidence Intervals Excluding N/As



As depicted in Table 3 and the forest plot above, no statistical significance was established for the exposure-outcome odds ratio, adjusted exposure-outcome odds ratio, any particular crude odds ratio under stratification, or adjusted odds ratios under stratification.

Discussion

Our statistically insignificant findings contrast with prior studies which have found associations between drinking and online sports betting. One cohort study measured alcohol use frequency in young sports bettors over one year and found that at the within-person level, there was a statistically significant association between drinking frequency and sports gambling engagement (Rate Ratio = 1.174, 95% CI: 1.109, 1.243) (Graupensperger et al. 2025). While multiple studies support an association between drinking and sports betting, the causality of the relationship is unclear. Huggett et al. describes inconsistencies among laboratory studies on whether alcohol use directly increases gambling behavior by altering the dopaminergic pathways in the brain, which some studies claim “augment reward learning, alter motivational processes, and heighten the salient aspects of gambling” (2019). Graupensperger et al. also discusses uncertainty about causality and clarifies that alcohol use frequency in their study was treated as a covariate for sports betting outcome rather than exposure (2025).

Our study controlled for multiple measures known to influence drinking and gambling behaviors, such as age, sex, household income, and peer influence. Although information bias is unavoidable when measuring variables via questionnaire, we tried reducing bias by phrasing questions objectively and assuring anonymity in the survey.

By nature of our cross-sectional study design, we were unable to determine temporality of the association between social drinking and online sports betting. Participants were selected from GroupMe chats of clubs we were involved in, thus sampling bias is a concern, and our results are not generalizable to the entire UNC undergraduate population due to convenience sampling. Since all data was collected via survey, both self-selection bias and information bias, specifically misclassification due to potential underreporting, are a concern in this study. Furthermore, our study had a small sample size which caused our study to be more vulnerable to such biases and produced high uncertainty and very wide confidence intervals. Our study also failed to measure certain confounders, such as mental health issues, which have been treated as covariates in similar studies.

We found no statistically significant association between drinking and online sports betting. Even after adjusting for covariates, the odds of online sports betting was not sufficiently different between those who participate in social drinking and those who do not. Our study analyses on potential confounders show that the association of drinking and online sports betting is modified by gender, household income, family and peer influence, opinion on sports betting at UNC, and perspective on the practice itself. Directionality of the measured association could not be determined due to our cross-sectional study design. Future work should conduct cohort studies with larger sample sizes to better capture the association and determine directionality between social drinking and online sports betting.

With the recent rise in internet and technology usage, more people have been exposed to online gambling advertisements, and online sports betting has only become more prevalent since the US Supreme Court allowed its state legalization in 2018 (Eisenberg et al. 2026). The National Council of Problem Gambling reported 75–80% of college students gambled, and

overall participation in fantasy sports gambling quadrupled from 2004 to 2018 (Winters & Derevensky, 2020). Multiple studies have also suggested sports betting is associated with gambling disorders, mental health issues, and binge drinking. As the popularity of online sports betting continues to grow, more research on this topic is needed to improve clinical detection for these conditions, inform the public on associated risks, and guide future policymaking to protect vulnerable individuals.

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